FORM TP 23243

TEST CODE **000574**

MAY/JUNE 2003

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

MATHEMATICS

UNIT 2 – PAPER 02

2½ hours

04 JUNE 2003 (p.m.)

This examination paper consists of THREE sections: Module 2.1, Module 2.2 and Module 2.3.

Each section consists of 2 questions.
The maximum mark for each section is 50.
The maximum mark for this examination is 150.
This examination consists of 6 printed pages.

INSTRUCTIONS TO CANDIDATES

1. **DO NOT** open this examination paper until instructed to do so.
2. Answer **ALL** questions from the **THREE** sections.
3. Unless otherwise stated in the question, all numerical answers **MUST** be given exactly **OR** to three significant figures as appropriate.

Examination material:

Mathematical formulae and tables
Electronic calculator
Ruler and graph paper

Section A (Module 2.1)

Answer BOTH questions.

1. (a) (i) Use the fact that $e^{-x} = \frac{1}{e^{x}}$ to show that

$$\frac{d}{dx}(e^{-x}) = -e^{-x}.$$

[3 marks]

(ii) Hence, evaluate

$$\int x^2 e^{-x}\,dx.$$

[4 marks]

(b) If $x = e^{-3t}\sin 3t$, show that

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 18x = 0.$$

[8 marks]

(c) (i) Given that $x = 2^y$, $y > 0$, express in terms of y

a) $\log_2 x$

b) $\log_x 2$

[4 marks]

(ii) Hence, or otherwise, solve the equation

$$\log_2 x = 8\log_x 2 + 2.$$

[6 marks]

Total 25 marks

2. (a) Express in partial fractions

$$\frac{1 - x^2}{x(x^2+1)}.$$

[8 marks]

(b) Use the substitution $u = \cos x$ to evaluate

$$\int \sin^3 x\,dx.$$

[7 marks]

2. (c) The rate of increase of a population of insects is directly proportional to the size of the population at time, t, given in days. Initially, the population is p_o and it doubles its size in 3 days.

(i) Show that $p = p_o e^{kt}$, where p is the size of the population after t days and $k = \frac{1}{3} \ln 2$. **[7 marks]**

(ii) Find the proportional increase in population at the end of

a) the first day

b) the second day. **[3 marks]**

Total 25 marks

Section B (Module 2.2)

Answer BOTH questions.

3. (a) Three sequences are given below.

$1,\ 4,\ 7,\ 10, \ldots$

$1,\ -\frac{1}{4},\ \frac{1}{7},\ -\frac{1}{10}, \ldots$

$(-1)^1,\ (-1)^4,\ (-1)^7,\ (-1)^{10} \ldots$

Determine which of the sequences is convergent, divergent or periodic and state which of the sequences is an arithmetic sequence. **[12 marks]**

(b) (i) Find the *nth* term of the series $1(2) + 2(5) + 3(8) + \ldots$ **[3 marks]**

(ii) Prove, by **Mathematical Induction**, that the sum to n terms of the series in (b) (i) above is $n^2(n + 1)$. **[10 marks]**

Total 25 marks

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4. (a) (i) Show that the series

$$\log_a b + \log_a (bc) + \log_a (bc^2) + \ldots\ldots + \log_a (bc^{n-1}),$$

where $a, b, c > 0, n \geq 1$

is an arithmetic progression whose sum, S_n, to n terms is $\frac{n}{2} \log_a b^2 c^{n-1}$.

[6 marks]

(ii) Find S_n when $n = 6$ and $a = b = c = 5$. **[2 marks]**

(b) Given the series $\frac{1}{2} + \frac{1}{2^4} + \frac{1}{2^7} + \frac{1}{2^{10}} + \ldots$

(i) show that the series is geometric **[3 marks]**

(ii) find the sum of the series to n terms **[3 marks]**

(iii) show that as n approaches infinity, the sum of the series approaches $\frac{4}{7}$.

[2 marks]

(c) (i) Write down, and simplify as far as possible, the FIRST THREE terms in the expansion of

$$(1 - ux)(2 + x)^4.$$

[7 marks]

(ii) Given that the coefficient of x^2 is 0, find

a) the value of u

b) the coefficient of x. **[2 marks]**

Total 25 marks

Section C (Module 2.3)

Answer BOTH questions.

5. (a) A bag contains 5 red balls, 7 black balls and 4 white balls. Two balls are drawn at random without replacement.

(i) Draw a tree diagram to represent the different ways in which the balls can be drawn from the bag, with branches showing the probabilities of drawing the red, black or white ball. **[5 marks]**

(ii) Find the probability of drawing 2 balls of the SAME colour. **[4 marks]**

(iii) Find the probability of drawing 1 red and 1 black ball. **[3 marks]**

(b) A journalist reporting on criminal cases classifies crime by age (in years) of the criminal and whether the crime is violent or non-violent. The table below shows that 150 criminal cases were classified in a particular year.

Type of Crime	**Age (in years)**		
	Under 20	**20 to 39**	**40 or older**
Violent	27	41	14
Non-violent	12	34	22

The journalist randomly selects a criminal case for reporting.

(i) What is the probability of selecting a case involving a violent crime? **[2 marks]**

(ii) What is the probability of selecting a case where the crime was committed by someone LESS than 40 years old? **[3 marks]**

(iii) What is the probability of selecting a case that involved a violent crime or an offender LESS than 20 years old? **[3 marks]**

(iv) Given that a violent crime is selected, what is the probability the crime was committed by a person UNDER 20 years old? **[2 marks]**

(v) Two crimes are selected for review by a judge. What is the probability BOTH are violent crimes? **[3 marks]**

Total 25 marks

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6. (a) Show that $e^{\ln y} = y$ for y real and positive. **[2 marks]**

(b) A disease spreads through an urban population.
At time, t, the proportion of the population who have the disease is p, where $0 < p < 1$.

The rate of change of p with respect to time is proportional to the product of p and $(1 - p)$.

(i) Form a differential equation in terms of p and t which models the situation described above. **[2 marks]**

(ii) Solve the differential equation. **[8 marks]**

(iii) Given that when $t = 0$, $p = \frac{1}{10}$ and when $t = 2$, $p = \frac{1}{5}$,

show that $\frac{9p}{1-p} = \left(\frac{3}{2}\right)^t$. **[10 marks]**

(iv) Find p when $t = 4$. **[3 marks]**

Total 25 marks

END OF TEST